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Education, Skills, and Wage Gaps in Canada and the United States

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ABSTRACT

We analyse data from the Programme for the International Assessment of Adult Competencies to reveal that immigrants in Canada and the United States make over \$200 less per month than native-born workers. In the United States, immigrants disproportionately work in low-wage occupations, leading to large mean national differences between immigrants and native workers. The wage differential disappears after accounting for education and cognitive skills, indicating policies must focus on reducing education and skill gaps in the United States. In Canada, an immigrant wage gap persists in nearly all occupational fields, suggesting that the better skilled and educated immigrants in Canada are not receiving the same wage premium as native workers. We close with implications for policy and future research.

INTRODUCTION

Canada and the United States have immigration policies that admit two very different pools of workers, yet in both countries immigrants consistently earn less than native-born employees (Aydemir and Sweetman, 2006; Batalova and Fix, 2015; Bloemraad and Provine, 2013; Coulombe et al., 2012; Nadeau and Seckin, 2010; Oreopoulos, 2011; Papademetriou and Sumption, 2011). Policy-makers argue that persistent immigrant wage gaps demonstrate that immigrants are not being fully – or equitably – integrated into national labour markets. For example, in 2006, the Canadian Minister of Citizenship and Immigration stated that immigrants’ “unemployment and underemployment represent more than just a drag on Canada’s productivity. It is a human tragedy, and basic decency dictates that it not be allowed to continue” (as cited in Wald and Fang, 2008, p. 458). Additionally, when first-generation immigrants are unable to find good jobs, their children are less likely to rise above the socioeconomic status of their parents (Zhou, 1997).

Some researchers suggest that national wage gaps exist because immigrants tend to work in low-wage occupations (Banerjee and Lee, 2015; Frank, 2011), and many scholars argue that additional education can help close earnings gaps (Banerjee and Lee, 2015; Barone and Van de Werhorst, 2011; Bowles and Gintis, 2002; Frank et al., 2013; Gintis, 1971). Yet there are competing explanations for why additional schooling may close wage gaps. For example, some scholars argue that education raises wages through “sheepskin effects”; that is, education is primarily valued in the labour market as a credential that signals a worker’s ability or personality traits and not because it increases skills (e.g. Banerjee and Lee, 2015; Bowles and Gintis, 2002; Gintis, 1971). Other scholars, however, conclude that education improves cognitive skills in ways that are valued by employers (e.g.,

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Baker, 2011, 2014; Baker et al., 2015; Barone and Van de Werhorst, 2011). In order to more fully understand why immigrants often earn less than native-born workers, we must consider how all three issues (occupational sorting, educational attainment, and cognitive skills) relate to immigrant wages.

In this article, we present a comparative study of Canada and the United States that examines whether immigrant workers are more likely to enter low-wage occupations and we demonstrate the effect of education on immigrants' skills and wages. Davies and Hammack (2005) noted that Canada and the United States "offer a strategic contrast" because the similarities of their schools "make it possible to isolate those factors responsible for any observed differences" in educational outcomes (p. 91). By choosing two countries that are geographically adjacent, comparably developed, and similar in terms of education systems, we are able to reach more valid conclusions about differences in wages between immigrant populations. We estimate models separately for both countries and compare the results of each model to determine whether the relationships between occupational fields, education, cognitive skills, and wages are similar in both labour markets. Thus, the comparative research design allows us to (a) examine how different immigration policies sort non-native workers into the labour market; (b) make more nuanced claims about the role of education in improving cognitive skills and, ultimately, wages; (c) test whether cognitive skills have independent, positive effects on immigrant wages in both countries. Two guiding questions direct our more detailed research questions:

1. Guiding and Research Question #1: Is there a wage gap between immigrants and native workers in the United States and Canada?
 2. Guiding Question #2: What are the potential factors driving the wage gaps in the United States and Canada?
- Research Question 2a: Do immigrants tend to work in lower-paying occupational fields?
 - Research Question 2b: After controlling for education and cognitive skills, do immigrant wage gaps exist in both countries within occupational fields?
 - Research Question 2c: Do differences in return to cognitive skills and educational attainment relate to the wage gap in both countries?

We use descriptive statistics, multinomial logistic regression models, and quasi-experimental path analysis models to answer these questions. We find that, on average, immigrant workers in the United States are concentrated in low-wage fields, but wage gaps disappear after controlling for education and cognitive skills. In Canada, immigrants do not disproportionately work in occupations that pay less, but wage gaps remain statistically significant after including education and cognitive skills in analyses. Our path models demonstrate that years of education (distinct from educational credentials) increases immigrants' cognitive skills in both countries.

These findings suggest that policymakers in both countries should consider labour market inequalities, occupational sorting, and educational opportunities if they want to integrate immigrant workers more fully into their societies. Canada's immigration system gives preferences to highly-educated immigrant workers, yet immigrants are paid less than native-born workers with similar skills and education. In the United States, immigrant workers tend to have equal earnings once schooling and education levels are taken into account. Thus, American policymakers should acknowledge that their wage gaps can be largely explained by immigrants being sorted into lower-status occupations which can, in part, be attributed to their lower cognitive skills and educational attainment. In both countries, schooling increases cognitive skills and earnings and can help close wage gaps.

In the next section we review existing literature and provide a theoretical foundation for our study. The third section introduces the data sources and methods used in this study. This is followed by findings which compare the results for each country. We close by proposing topics for future research and discussing implications for theory and policy.

REVIEW OF THE LITERATURE AND THEORETICAL FRAMEWORK

Over half of all immigrants to countries in the Organization for Economic Co-operation and Development (OECD) move to Canada or the United States. The proportion is even higher for OECD immigrants with tertiary education – approximately two-thirds choose to live in Canada or the United States (Grogger and Hanson, 2011). Although Canadian and American policies yield two different groups of workers in terms of education and skills, immigrants in both countries tend to earn less than their native-born peers (Aydemir and Sweetman, 2006; Bloemraad and Provine, 2013; Chiswick and Miller, 1990; Coulombe et al., 2012; Nadeau and Seckin, 2010; Oreopoulos, 2011; Papademetriou and Sumption, 2011). This study compares two countries that are otherwise similar (except in the ways they admit immigrant workers) to tease out the structural and individual reasons underlying immigrant wage gaps.

Some researchers suggest that immigrants earn less than native-born workers because they work in lower-paying occupations (Banerjee and Lee, 2015; Frank, 2011). Historically, the United States has used immigration policy to supply low-wage agricultural workers (Hanson, 2009; Massey, 1987). In contrast, one might expect that because Canadian immigrants tend to be highly educated, they would be unlikely to earn less than native-born workers. However, Canadian immigrants' education and cognitive skills do not guarantee employment in a desired occupation. Results from a regression decomposition analysis suggest that Canadian immigrants who aspire to work in high-status occupations are nearly twenty-five per cent less likely to work in their desired field. Conversely, immigrants who intend to work in lower-status occupations are much more likely to find employment in their fields of interest (Frank et al., 2011).

More often, scholars question whether immigrants earn less than native-born workers due to differences in education and skills (e.g. Bartalova and Fix, 2015; Borjas, 1994; Chiswick and Miller, 1990; Frank et al., 2013; Ono and Zavodny, 2008; Psacharopoulos and Patrinos, 2004). The literature on immigrants' education and skills tends to fall into two theoretical camps. The first camp suggests that schooling acts, primarily, as a credentialing process that improves wages through "sheepskin effects" (Banerjee and Lee, 2015; Bowles and Gintis, 2002; Gintis, 1971). A recent study suggests that even though immigrants to Canada tend to be more highly educated than native-born workers, employers may undervalue their foreign education credentials (Banerjee and Lee, 2015). Banerjee and Lee (2015) use longitudinal growth-curve techniques to find that additional schooling closes wage gaps, but they do not consider whether education actually improves immigrants' skills or whether measures of immigrants' skills are related to higher wages. Instead, they argue that immigrants who earn Canadian education credentials provide a "credible, well-understood signal to employers" who "then reward the immigrants commensurate to their productivity signal, leading to higher wage growth" (p. 212). This body of research implies that immigrants are being coerced into paying and working hard to collect educational credentials to certify productivity without necessarily improving their skills.

Other studies adopt a human capital orientation and find that wage gaps are related to true differences in skills and abilities (Barone and Van de Werhorst, 2011; Frank, 2011; Frank et al., 2013). Through the lens of human capital theory, employers set compensation levels based on the skills that immigrants bring and apply in the workplace. Thus, it may seem reasonable that workers with limited language fluency often earn less than native-born peers (Chiswick and Miller, 1990; Frank et al., 2013). However, research is often limited by a lack of valid measures of cognitive abilities. Even Frank and colleagues (2013) admit that their measure of language proficiency is "indirect" but preferred to the "unsatisfactory" measure collected by the Canadian census. Because researchers have limited data, it is difficult for them to test whether schooling has an effect on immigrants' cognitive abilities and whether the two, together, are related to wages.

Previous studies tend to focus on education and skills as separate sources of human capital (or, alternately, stratification), but a series of new sociological studies suggest that education and skills

are complementary and work together to transform the nature of occupations and occupational credentialing (Baker, 2009; 2011; 2014; Baker et al., 2015). Through a series of experiments, Baker and colleagues (2015) demonstrate that schooling improves cognitive abilities such as numeracy. At the same time, educational credentials are now the most legitimate form of matching workers to occupations (Baker, 2011; 2014). From this perspective, education has revolutionized society in such a way that it would be nonsensical to view wages as the result of *either* education *or* cognitive skills. Instead, we must consider whether wages are a function of *both* education *and* cognitive skills. Few studies have conducted these types of analyses because they would require large-scale data sets with accurate measures of immigrants' occupational fields, education, and cognitive skills.

This brief overview of research literature helps contextualize the two (related) rationales for why immigrant wage gaps exist. On one hand, we might understand wage gaps as resulting from national-level processes: immigrants are admitted according to government policies that favour certain types of workers, and then immigrants are sorted in the labour market into lower- or higher-paying occupations. On the other hand, wage gaps can result at the individual level due to workers' education and skills. We expect to find that in both countries education has positive, significant effects on immigrants' cognitive skills – and that the two together are related to higher wages. In the section that follows, we introduce a new dataset that allows us to answer our research questions and to contribute to the literature on immigrants' wages, skills, and education.

METHODOLOGY

Data and variables

Data from the 2012 Program for the International Assessment of Adult Competencies (PIAAC) are used in this study. PIAAC is a large-scale, cross-national adaptive assessment that tests individuals' literacy and numeracy skills. Respondents complete different portions of the assessments, so ten plausible values are generated by PIAAC staff to estimate participants' scores on the complete range of literacy and numeracy test items. Additionally, demographic and occupational information is collected through background questionnaires. PIAAC includes nationally representative samples of workers aged 16–65 from both countries (OECD, 2013).

To examine our research questions, first-generation immigrants were compared with individuals who were born in Canada or the United States, respectively. (Second generation immigrants are omitted from this study due to small sample size. To avoid misinterpretation they are not included in the native-born category.) Cases missing information on immigrant status or wages were omitted from this study. Additionally, wage outliers, which we define as individuals whose monthly wage is greater than three standard deviations above the mean wage for their occupational field, were deleted. Multiple imputation was used to address problems with missing data for the remaining control variables (Wayman, 2003). Our final sample for the US study was approximately 2,700 of which 12.8 per cent were first generation immigrants. The final sample for Canada included approximately 15,660 participants of which 17.9 per cent were first generation immigrants. Table 1 provides a breakdown of all included variables by immigrant status.

To calculate the wage gap between first generation immigrants and their native peers, we used self-reported monthly *wage* plus bonuses in US dollars and adjusted for inflation as the dependent variable (results were also verified using logged wage values). Primary explanatory or independent variables include years of education and literacy and numeracy skills. *Years of education* is derived from PIAAC data identifying an individual's highest educational credential, using the International Standard Classification of Education (ISCED) scale. Similar to Byun and Kim (2010), *years of education* is recoded from ISCED categories using country specific mapping information from

TABLE 1
DESCRIPTIVE STATISTICS FOR CANADA AND UNITED STATES SAMPLES

	Canada			United States		
	Overall Sample	First Generation Immigrant	Native	Overall Sample	First Generation Immigrant	Native
Percent of Sample	100%	17.88%	82.12%	100%	12.84%	87.16%
Monthly Earnings, USD PPP	3398.77 (2511.77)	3207.44 (2557.59)	3440.44 (2499.84)	3964.64 (3511.57)	3718.56 (3432.19)	4000.89 (3522.37)
Literacy	274.45 (45.41)	263.57 (49.72)	276.82 (44.05)	277.46 (45.11)	244.91 (53.88)	282.26 (41.58)
Numeracy	265.8 (49.63)	258.29 (55.62)	267.44 (48.08)	262.62 (51.79)	233.99 (64.04)	266.84 (48.34)
Years of Schooling	13.34 (2.50)	14.66 (1.75)	13.05 (2.54)	13.72 (2.79)	13.11 (3.95)	13.81 (2.56)
ICT Skills	92.98% (0.26)	94.29% (0.23)	92.86% (0.26)	93.90% (0.24)	91.07% (0.29)	94.31% (0.23)
Age	40.42 (13.02)	39.86 (11.68)	40.54 (13.29)	40.02 (13.07)	39.46 (11.77)	40.10 (13.25)
Female	52.22% (0.50)	50.70% (0.50)	52.55% (0.50)	52.68% (0.50)	47.84% (0.50)	53.40% (0.50)
Hours Worked Per Week	37.38 (13.02)	36.94 (11.99)	37.47 (13.23)	39.7 (13.28)	40.16 (12.44)	39.63 (13.41)
Years in Position	7.99 (9.12)	5.16 (6.85)	8.60 (9.43)	7.42 (8.35)	6.10 (6.78)	7.61 (8.54)
Arrived in Country before Age 20	NA	31.82% (0.47)	NA	NA	47.69% (0.50)	NA
<i>n</i>	15660	2800	12860	2700	350	2350

Note: Standard deviations reported in parentheses.

Appendix 5 of the PIAAC technical report (e.g. in the US completion of ISCED 1 = 6 years of education and completion of ISCED 6 = 21 years of education). Literacy skills and numeracy skills are identified through PIAAC's *literacy* and *numeracy* assessments. Literacy and numeracy scores range from 0 to 500. The inclusion of literacy and numeracy skills in the analyses allows us to parse out differences between returns to individual cognitive skills or abilities and returns to years of schooling. To help ensure that numeracy was not substantially affected by language proficiency, participants in Canada were given the option of having the test administered in either English or French. According to the PIAAC technical report, the multiple language options offered in Canada captured the native language of 98.4 per cent of the sampling frame. The US had a monolingual (English) requirement, which only covered 91.5 per cent of the sampling frame, suggesting that a larger percentage of participants in the US were likely to complete the literacy and numeracy assessment in a language other than their native language (OECD, 2013).

Beyond education and literacy and numeracy skills, factors that may affect an individual's wage include the experience they have in the position, the number of hours per week they work, their area of study, and their adeptness with information communication technologies (ICT). To address the confounding nature of these factors we treated each factor as a control variable in this study. *Hours worked per week* and years in the position are continuous variables. *Years in the position* is derived by subtracting an individual's age from the age they started their present position. Identifying an individual's area of study is important because individuals who majored in engineering, for example, may have higher numeracy scores that may lead to higher wages. To control for *area of*

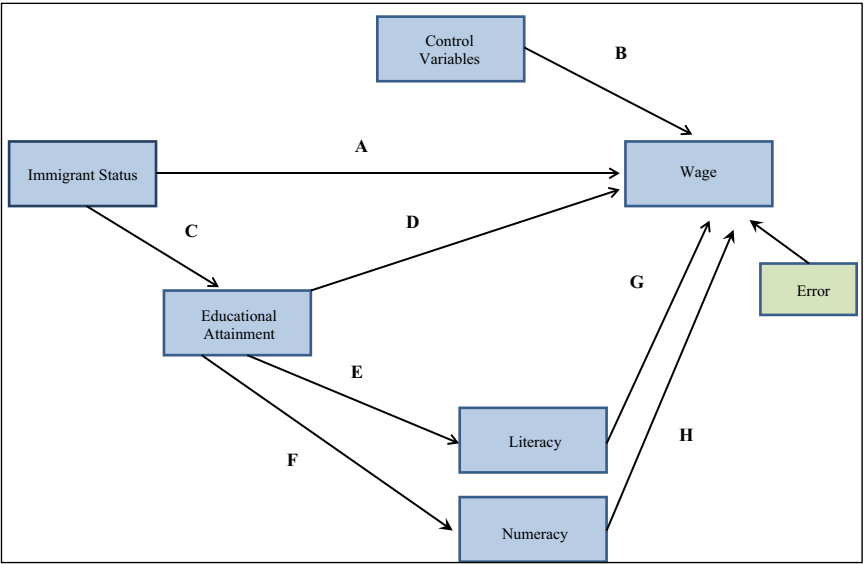
study, we create dummy variables from eight broad educational categories: services or general education, education-focused, humanities, social sciences, science or math, engineering, agriculture or veterinary medicine, and health. *ICT* adeptness is a context specific measure that asks the individual whether they have the ICT skills necessary to be successful in their occupation (1 = yes, 0 = no). Finally, *gender* (female = 1, male = 0) and *age* (continuous variable) are included to control for discrepancies in wage by gender and experience.

Methods of Analysis

To answer the first research question (Do immigrants tend to work in lower-paying occupational fields?), we used multinomial logistic estimation to predict the occupational categories of all the respondents in each country based solely on their immigrant status. We used path analysis to answer the second research question (After controlling for education and cognitive skills, do immigrant wage gaps exist in both countries within occupational fields?) and the third research question (Is education positively related to cognitive skills and wages, and are cognitive skills related to higher wages?). Path analysis, also called simultaneous regression (Lleras, 2005), allows us to distinguish between (a) the direct effect of immigrant status on individual wage and (b) the indirect effects of immigrant status on wage through differences in years of education and literacy and numeracy skills.

The complete path analysis model was finished through four additive steps with the complete model shown in Figure 1. The baseline model predicts the direct effect of immigrant status on individual wage (path A) for each occupational field, by country (note that in the US sample, there were too few immigrants working in some occupational fields to complete path analyses for all eight occupational groups; see additional explanation in the next section). The second model adds control variables (path B) – years in the position, hours worked per week, area of study, ICT adeptness, gender, and age – to the baseline model. The third model also includes years of education

FIGURE 1
COMPLETE PATH MODEL



(paths C and D), which captures the indirect effect of immigrant status through years of education. The final model adds literacy (paths E and G) and numeracy skills (paths F and H) as an outcome of years of education, allowing us to distinguish between credential effects and skill premiums on wage.

Akaike Information Criteria (AIC) and Bayesian Information Criteria (BIC) are used to assess which model best fit the available data. AIC and BIC measure the overall model fit for both nested and non-nested models, identifying how likely the model is to generate the original data given the specified parameters. For AIC, smaller numbers indicate desirable model fit; for BIC, the larger negative number indicates the preferred model (Long and Freese, 2001). Given the use of plausible values in the measurement of literacy and numeracy skills, all analyses that include measures of cognitive abilities are estimated ten times, once for each plausible value, and are combined using Rubin's (1987) rule to provide appropriate standard errors. Analyses are conducted independently by country, allowing us to compare path coefficients between countries and evaluate the relative strength of explanatory variables, and use the final weight provided by PIAAC.¹

Additionally, to ensure that the immigrant-native wage differential is not merely an indicator of occupational difference, separate within-country path analyses are conducted for each occupational field. Occupational categories are based on the first level of the 2008 International Standard Classification of Occupations (ISCO-08). Two groups, "Skilled Agricultural, Forestry, and Fishery Workers" and "Armed Forces Occupations" are omitted from the sub-analysis of both countries due to small sample size. See Appendix C in Smith and Fernandez (2015) for a breakdown of occupational fields.

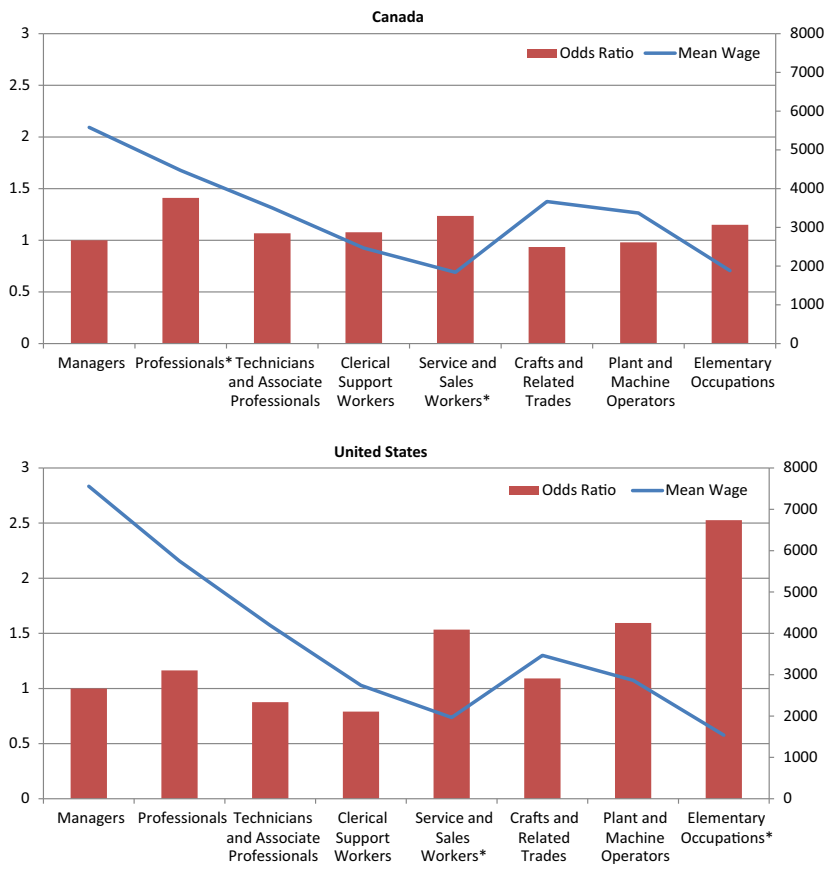
RESULTS

Bivariate analysis reveals the presence of initial wage gaps between immigrants and native workers in both the United States and Canada. In Canada, immigrants make, on average, \$233.00 less per month than native workers ($p < .001$) while in the United States the difference stands at \$282.33 in favor of native workers. The gap in the United States, however, does not meet the standard of statistical significance ($p = .162$). This may be due to the smaller sample size in the United States and the challenges of including the large undocumented immigrant population in national samples (Smith, 2012). Given that undocumented immigrants in the United States are more likely to have lower wage jobs (Passel, 2006), their omission may artificially compress the difference in wages and make a larger p -value more likely.

The initial wage gap in each country could be a product of a disproportionate concentration of immigrants in lower wage occupations. This may especially be the case in the United States, given the education ($-.71$, $p < .001$) and skill gaps (Literacy = 37.35, $p < .001$; Numeracy = 32.85, $p < .001$) between immigrants and natives.² Figure 2 reports results from the pair of null multinomial logistic regression models that predict immigrants' occupational fields in each country.

In Canada, immigrants are not concentrated in lower wage fields. Overall odds ratios hover around the reference group (managers; see Figure 2). Compared to the reference group, immigrants are statistically significantly more likely to hold positions in the lower wage service and sales field as well as the higher wage professional field. However, immigrants in the United States are more likely to be in the two lowest wage occupational fields, service and sales workers and elementary occupations. In the United States, the odds of an immigrant holding a position in the elementary occupation field is over 2.5 times greater than that of a native. In other words, relative to their native peers, immigrants in the United States are over two and a half times more likely to hold positions in the lowest wage elementary occupations field. The greater concentration of immigrants in the two lowest wage fields helps explain the initial immigrant wage gap present in the United States.

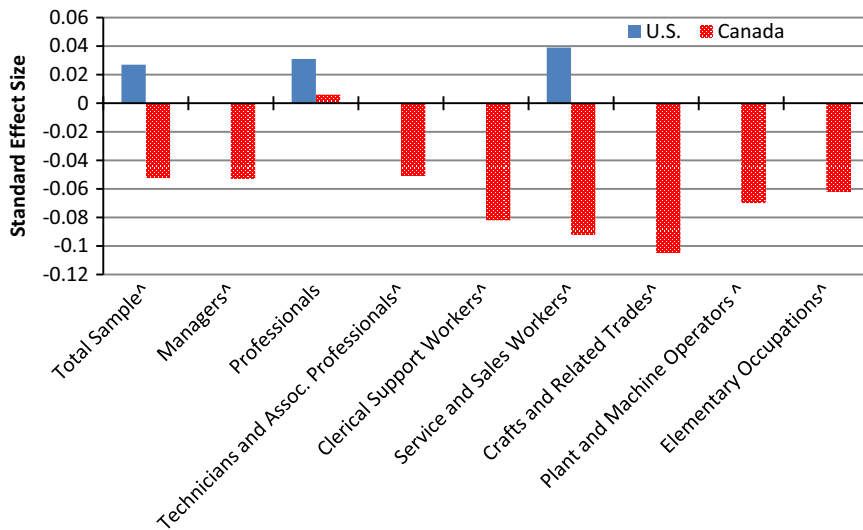
FIGURE 2
IMMIGRANT WORKERS' ODDS OF BEING EMPLOYED IN OCCUPATION FIELDS IN CANADA AND THE UNITED STATES



Note: Managers were used as the reference group (OR=1.0). * $p < .05$.

Because the immigrant wage gap in Canada cannot be explained by a greater concentration of immigrants in lower wage occupations, it is important to examine whether the wage gap is a by-product of large wage differences within occupational fields. To address the second research question (After controlling for education and cognitive skills, do immigrant wage gaps exist in both countries within occupational fields?), we use additive path analyses for each occupational field, by country. In Figure 3, we report the standardized effect of immigrant status on *wage* after controlling for education, cognitive skills, and control variables in various occupational fields (path A in the full model). As immigrants in the US are disproportionately found in low wage fields, the sample size of immigrants in some occupational field are insufficient to run the full model. Using a standard of five observations per estimated parameter (Bentler and Chou, 1987; Bollen, 1989), analyses were not performed for occupational fields in the United States with less than 50 immigrants included in the sample (see appendix A for sample sizes used in

FIGURE 3
RESULTS OF PATH ANALYSIS MODELS TESTING FOR STATISTICALLY SIGNIFICANT IMMIGRANT WAGE GAPS WITHIN OCCUPATIONAL FIELDS, BY COUNTRY (CONTROLLING FOR EDUCATION, COGNITIVE SKILLS, AND BACKGROUND CHARACTERISTICS)



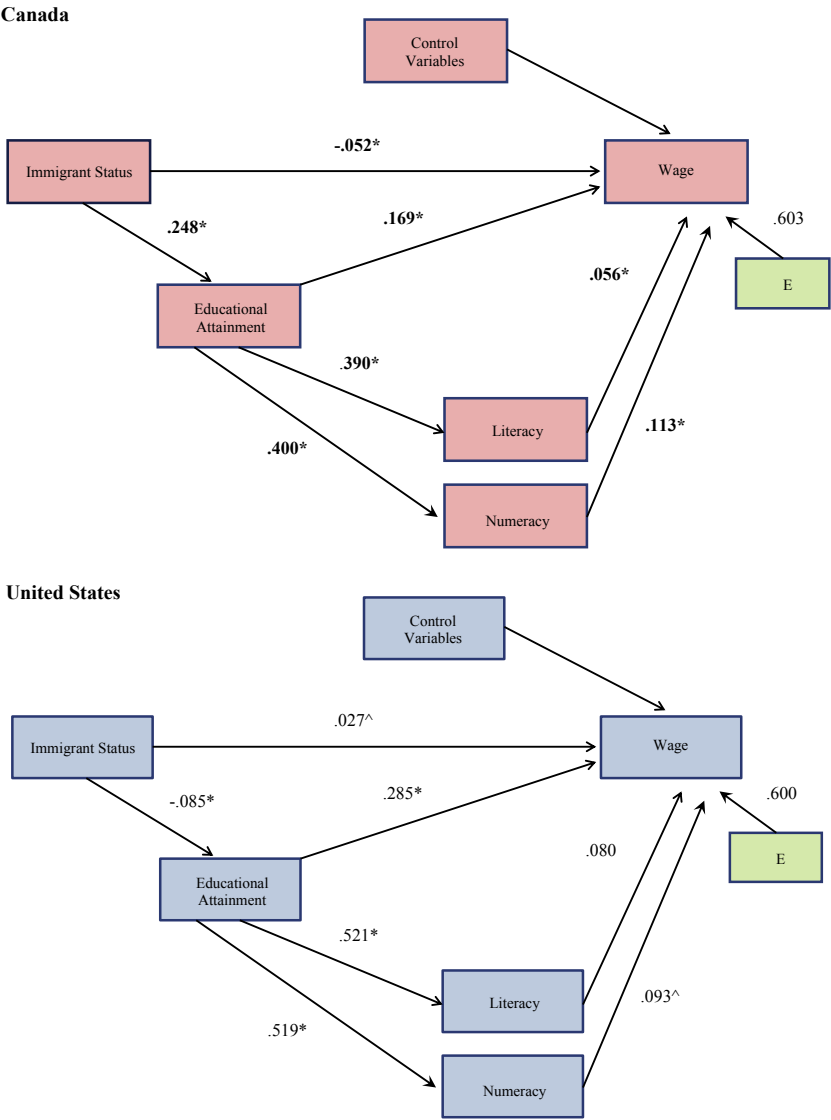
Note: Standardized coefficients provided. Negative numbers indicate immigrant wage gap. Positive numbers indicate immigrant wage advantage. ^ $p < .05$ (Canada).

all models). In the United States sample sizes were only large enough to complete path analyses for the total sample and for workers in the “professionals” and “service and sales workers” occupational groups.

Immigrant wage gaps (i.e. the direct effect of immigrant status on wage) are found in every occupational field in Canada, sans professionals. These gaps increase in magnitude once education and literacy and numeracy skills are included in the model. This contrasts sharply with the United States where, once demographic control variables are included in the model, wage gaps in the professionals and service and sales workers occupational groups are eliminated. In addition to a disproportionate concentration of immigrants in low wage jobs and unequal distributions within occupational fields, differential returns on educational attainment and skills may help explain the immigrant wage gaps in the United States and Canada. Figure 4 displays the complete path analysis model used to explore this potential factor, for the full samples of workers in each country.

In Figure 4, path A indicates that after accounting for education, literacy and numeracy skills, and other control variables, there is a negative relationship between immigrant status and wages in Canada. Conversely, in the United States, when immigrants and native-born workers are similarly educated and skilled, immigrant status is positively related to earnings (although marginally significant at the $p < .10$ level). Furthermore, in Canada, immigrants tend to have higher levels of educational attainment, while in the United States, immigrant status is negatively related to years of schooling (path C). Next, path D shows the main effect of years of education on wages in both Canada and the United States. In both countries, education has the strongest positive effect on wages. Returns to schooling tend to be greater in the United States, regardless of immigrant status.

FIGURE 4
PATH ANALYSIS MODELS USED TO TEST MAIN AND INTERACTIVE EFFECTS OF IMMIGRANT STATUS, YEARS OF EDUCATION, AND COGNITIVE SKILLS ON EARNINGS



Note: Standardized coefficients provided. *indicates $p < .05$. ^ indicates $p < .10$.

The two arrows in the bottom, left-hand corner of Figure 4 (paths E and F) illustrate the main effects of education on workers' literacy and numeracy skills. In both countries, education appears to have positive relationships with respondents' cognitive abilities. Looking at paths G and H, we see that cognitive skills have direct effects on wages while education has indirect effects on

workers' wages through cognitive skills. Numeracy skills have greater returns than literacy skills in Canada, but literacy skills do not have a statistically significant relationship with wages in the United States. The inclusion of skills and years of education allowed us to segment out return on schooling compared to the return on skills.

DISCUSSION AND CONCLUSION

Findings from this study support previous studies, which find that immigrant workers make significantly less than their native peers. In the United States and Canada, a similar initial immigrant wage gap is present despite their strikingly different immigrant populations. Once disaggregated by occupational field, wage gap patterns remain stable within each country. After controlling for education, numeracy and literacy skills, as well as a host of other demographic and work specific variables, a significant immigrant wage gap is present in nearly all occupational fields in Canada (with the exception of professionals) with the monthly wage differential ranging from \$223 to \$601. However, in the United States there is no evidence of an immigrant wage gap in the professionals and service and sales workers occupational groups after accounting for education, cognitive skills, and a vector of control variables. Our findings suggest that within these occupations immigrants to the United States make roughly the same monthly wage as their native peers with equivalent education and literacy and numeracy skills.

A more in-depth look at the immigrant wage gap by country revealed different potential explanations which guide our policy recommendations. In the United States, the initial wage gap appears to be an artifact of the disproportionate concentration of immigrants in low-wage positions. The disproportionate concentration suggests that immigrants in the United States, in general, are having difficulty accessing higher wage jobs. This may be due in part to immigrants' significantly lower levels of education and literacy and numeracy skills which has been well documented in past research (Bartalova and Fix, 2015; Borjas, 1994; Chiswick and Miller, 1990; Frank et al., 2013; Ono and Zavodny, 2008; Psacharopoulos and Patrinos, 2004; Smith et al., 2015). Given that nearly 50 per cent of immigrants in the United States arrived before the age of 20, many immigrants complete their education in American schools. Only 2 per cent of first generation immigrants arrive to the United States with foreign education qualifications. Therefore, to address the immigrant wage gap in the United States, policy-makers should focus on the well-documented educational attainment and achievement gaps between immigrants and natives (e.g. Harris et al., 2008; Portes and Rumbaut, 2001). Policies that aid persistence by supporting the transition of immigrants into the American education system and train educators to support learners who are culturally and linguistically diverse are essential to reduce the wage gap. Closing the attainment gap is especially crucial given the high wage premium placed on education credentials in the United States (relative to cognitive skills) regardless of immigrant status.

The persistence of an immigrant wage gap in Canada across nearly all occupational fields after controlling for education, literacy skills, numeracy skills, age, gender, years in the position, hours worked per week, area of study, and ICT adeptness is interesting and requires further study. Our evidence suggests that the wage gaps in Canada are not driven by differences in cognitive skills. One potential contributor to the gap is the differential return on education and skills in Canada, relative to the United States. Literature suggests that because immigrants in Canada have completed significantly more years of education than their peer counterparts, part of the wage gap may be due to an unequal education premium disadvantaging immigrants (e.g. Coulombe et al., 2012; Nadeau & Seckin, 2010; Oreopoulos, 2011). Additional research using wage decomposition analysis is needed (such as Frank, 2011).

The issues and potential solutions to the immigrant wage gap in Canada result largely from their point-based immigration policy, which has successfully attracted higher skilled and more educated immigrants than the family reunification focused policy of the United States. Given the education advantage of

immigrants in Canada and fact that over 68 per cent of the immigrants in the Canadian sample arrived after the age of 20, solutions should be targeted at larger immigration and economic policies.

Our results suggest that at least three factors contribute to the immigrant wage gap in Canada. First, immigrants in Canada are underemployed. There is little difference in the years of education completed by immigrants regardless of their occupational field (see also Smith and Fernandez, 2015), suggesting that many immigrants in Canada have educational credentials well above that generally required in their occupation. Second, the limited return on education in Canada disproportionately plagues immigrants whose significantly greater levels of education should lead to economic advantage. Third, the limited return on education may be due to international education credentials signaling different information to employers in Canada. As the majority of immigrants in Canada arrive after the age of 20, many have completed their education in their countries of origin. Specifically, approximately 50 per cent of first generation immigrants arrive to Canada with a foreign educational qualification beyond that of secondary school completion. International credentials may be downgraded in the economy due to perceived discrepancies with Canadian credentials. The education advantage for immigrants coupled with their persistent skills gap suggests that this signal may have some merit. This result complements prior work that suggests there may be a sheepskin effect for merely having a degree or completing certain education level that is more pronounced for immigrants (Batalova and Fix, 2015), especially in Canada (Banerjee and Lee, 2015).

It is important to recognize that the wage gap in Canada is not a by-product of controlling for education – which essentially compares natives and immigrants with similar education and eliminates the education advantage from the wage gap equation – but is present before adjusting for education. Controlling for the immigrant education advantage in Canada does not expose the immigrant wage gap but expands it. The presence of a wage gap across nearly all occupational fields in Canada suggests that there may be underlying cultural preferences or practices that lead to differential wage outcomes. The presence of differential wage returns across occupational fields in the country leads to policy recommendations centred on diversity training and equitable hiring procedures. Additionally, continuing education for adult immigrants may help immigrants align their literacy and numeracy skills with their education credentials and eliminate the undervalued signal from their international credential. Finally, although the point-based immigration policy appears to be successful at recruiting highly educated immigrants, more needs to be done to support their transition once they arrive in Canada. For example, occupational placement services may alleviate the mismatch between qualifications and occupational opportunities.

In summary, while an initial immigrant wage gap is present in both the United States and Canada, the differential wages appear to be influenced by unique factors. For the United States, the well documented education and skills gaps leads to occupational sorting with a disproportionate number of immigrants in low wage positions. After controlling for education and cognitive skills, the wage gaps disappear, suggesting that immigrants and native workers with equal skills and levels of education make, on average, the same wage. For Canada, the immigrant wage gap is not the result of unequal occupational sorting, instead there are statistically significant differences in earnings across the majority of occupational fields. The persistence of wage gaps in nearly every occupation is due, in part, to the lower premium for skills in Canada and the disproportionate return on education for immigrants. This suggests that there may be strong sheepskin effects connected to the Canadian education credential in Canada. Conversely, in the United States wage returns appear to be driven to a relatively greater extent by skills, supporting the human capital hypothesis.

Limitations and future research

This comparative study provides important insights into immigrant wage gaps in Canada and the United States; however, the study has limitations that can direct future research. The differential

return on skills and education leads to potential research that utilizes matching approaches, such as propensity score matching, to estimate whether the immigrant wage gap in Canada is indeed a systematic issue or dependent on the unique characteristics of immigrants in Canada. In addition, conclusions regarding differences between international and national education credentials are preliminary and based on immigrant age of arrival. Incorporating information specific to where the highest education credential was obtained may corroborate and extend our findings.

The small immigrant sample in the United States limited the depth of the investigation. Although the variance in wage is greater between the large occupational fields than within them, we cannot rule out heterogeneity of wage in occupational fields as a contributing factor to wage gaps in Canada. The potential within-field variance may help explain divergent findings with other research that finds wage gaps within professional fields in Canada (Reitz, 2001). This could be addressed by expanding the occupational identifier to the second level ISCO-08 code. However, because of the small sample in the United States, further disaggregating occupational groups would reduce the power needed to complete our analyses. Future studies could focus on heterogeneity within occupational fields in Canada.

Additionally, our study was limited in the ways we considered wage differences between women and men. For example, among immigrant populations women tend to be concentrated in low-wage occupations and paid less than men working in the same field (e.g. Beach and Worswick, 1993; Tastsoglou and Preston, 2012). In our study we did not conduct subgroup analysis to examine wage differences between men and women. However, to account for wage gaps between men and women, we added a dichotomous gender control variable in our path models.

Immigrant wage differences may also have been related to the amount of time that elapsed since the worker first entered Canada or the United States. Our results may have been biased by not including a variable to control for the number of years since the respondents relocated to their new host countries. However, we included several other variables that served as proxies including “age,” “arrived in country before age 20,” and “years in the position.” In the future, researchers may try to collect original data or conduct secondary data analyses that use more precise measures of the number of years that passed since immigrants entered a country with a new educational system and labour market.

Finally, as outlined in past research, country of origin plays an important role in the experience of immigrants (Bertrand and Mullainathan, 2003; Jacquemet and Yannelis, 2012; Oreopoulos, 2011). Unfortunately, due to the small United States sample of immigrants, we were not able to include country of origin in our analysis. Clarke and Skuterud (2016), analysing data from the Adult Literacy and Lifeskills (ALL) survey, found that there is a bimodal distribution of immigrants in the United States, with approximately 40 per cent having a Spanish-speaking background and 40 per cent coming from countries where Spanish is not the dominant language; the latter group tend to be more highly educated than Americans who were born in the United States. Clarke and Skuterud found that, when it comes to literacy skills, the non-Spanish-speaking immigrant group in the United States was similar to the immigrant sample in Canada. However, in Canada and the United States different immigration policies result in different immigrant populations. As previously mentioned, the United States has a substantially different immigrant population from Canada, in part due to the United States prioritizing family reunification. We argue that it is necessary to include all immigrants in our analyses, regardless of ethnic or language background, to fully understand the education and wage differentials between groups.

Our analyses also move beyond literacy as the sole indicator of cognitive skills to include a measure of respondents’ numeracy skills, which were shown to be statistically significantly (a) moderated by immigrant status and (b) related to monthly wages. The numeracy tests should be less related to countries of origin because, even though literacy may be considered an “enabling factor for numerate behavior,” the PIAAC numeracy test was designed to assess “a wide range of skills and knowledge” and not simply the application of “arithmetical skills to information embedded in

text” (OECD, 2013, p. 3). In other words, the numeracy and literacy tests were designed to assess different domains of cognitive skills so that the numeracy test is a valid measure of numerate cognition and therefore more appropriate in comparing seemingly different immigrant populations. Thus, our inclusion of numeracy skills expands on work by Clarke and Skuterud (2016) and shows that multiple measures of cognition are related to immigrant status as well as immigrants’ wages. Future quantitative research should continue to use multiple skill variables and attempt to disaggregate immigrant samples to test whether the effect of immigrants’ numeracy skills varies by country of origin.

Finally, we specified models that tested whether education affects cognition, and whether the two together affect wages. However, some may argue that education and cognition are endogenous, in other words, scholars may suggest that people with higher literacy or numeracy skills are more likely to complete additional years of schooling. Our theoretical framework was based on an experimental design, which showed that education affects cognitive skills (Baker et al., 2015). Thus, we modelled our path analyses uni-directionally from education to skills – and ultimately to wages. Although we specified our models based on our theoretical framework, future research may continue to explore the direction of the causal relationship between cognition and education.

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NOTES

1. Additional analyses applying the 80 replicate weights provided by PIAAC did not substantially change the magnitude of the coefficient and had no effect on the significance level. Full results with replicate weights are available from the authors upon request.
2. Immigrants in Canada have an educational attainment advantage (1.61, $p < .001$) relative to native workers while the skills gap is only $1/3^{\text{rd}}$ that of immigrants in the United States (Literacy = 13.25, $p < .001$; Numeracy, $p < .001$).

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APPENDIX A

SAMPLE SIZES FOR REGRESSION MODELS

	Canada (n)	United States (n)
Multinomial Regression Models		
Immigrant Sample	2800	350
Path Analysis Models		
Total Sample	12860	2350
Managers	1480	260
Professionals	3350	620
Technicians and Associate Professionals	2980	450
Clerical Support Workers	1350	220
Crafts and Related Trades	1300	200
Plant and Machine Operators	990	170
Elementary Occupations	1220	180
Service and Sales	2820	580

Note: Samples are rounded due to concerns about identification of immigrant individuals in the US given the number of parameters and relatively small sample size.